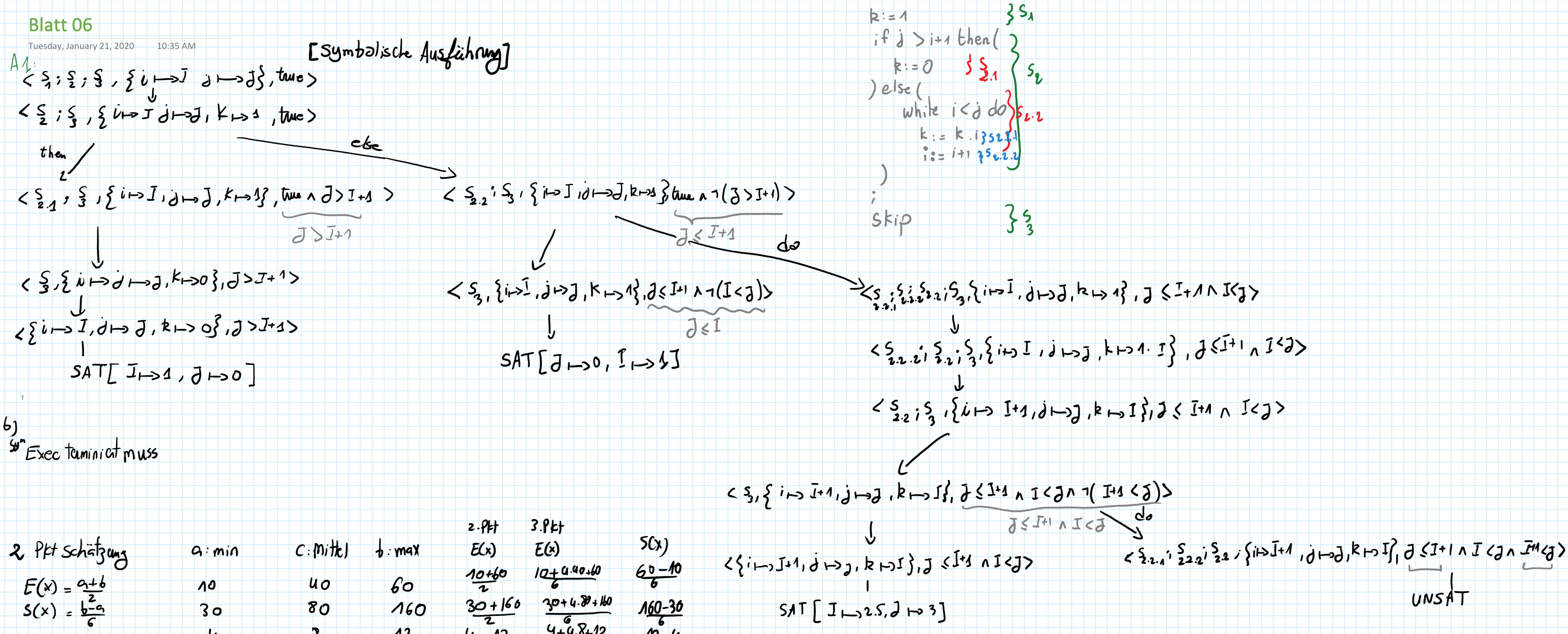
Blatt 06

Tuesday, January 21, 2020 10:35 AM

A1:

[Symbolische Ausführung]



```
k:=1                } S1
if j > i+1 then(    } S2
   k:=0             } S2.1
) else (
   while i<j do     } S2.2
     k := k.i       } S2.2.1
     i:= i+1        } S2.2.2
)
;
skip                } S3
```

$\langle S_1; S_2; S_3, \{i \mapsto I, j \mapsto J\}, true\rangle$

$\downarrow$

$\langle S_2; S_3, \{i \mapsto I, j \mapsto J, k \mapsto 1\}, true\rangle$

then

$\langle S_{2.1}; S_3, \{i \mapsto I, j \mapsto J, k \mapsto 1\}, \underbrace{true \wedge J > I+1}_{J > I+1}\rangle$

$\downarrow$

$\langle S_3, \{i \mapsto j \mapsto J, k \mapsto 0\}, J > I+1\rangle$

$\downarrow$

$\langle \{i \mapsto I, j \mapsto J, k \mapsto 0\}, J > I+1\rangle$

SAT[$I \mapsto 1, J \mapsto 0$]

else

$\langle S_{2.2}; S_3, \{i \mapsto I, j \mapsto J, k \mapsto 1\}, \underbrace{true \wedge \neg(J > I+1)}_{J \leq I+1}\rangle$

$\langle S_3, \{i \mapsto I, j \mapsto J, k \mapsto 1\}, \underbrace{J \leq I+1 \wedge \neg(I<J)}_{J \leq I}\rangle$

$\downarrow$

SAT[$J \mapsto 0, I \mapsto 1$]

do

$\langle S_{2.2.1}; S_{2.2.2}; S_{2.2}; S_3, \{i \mapsto I, j \mapsto J, k \mapsto 1\}, J \leq I+1 \wedge I<J\rangle$

$\downarrow$

$\langle S_{2.2.2}; S_{2.2}; S_3, \{i \mapsto I, j \mapsto J, k \mapsto 1 \cdot I\}, J \leq I+1 \wedge I<J\rangle$

$\downarrow$

$\langle S_{2.2}; S_3, \{i \mapsto I+1, j \mapsto J, k \mapsto I\}, J \leq I+1 \wedge I<J\rangle$

$\langle S_3, \{i \mapsto I+1, j \mapsto J, k \mapsto I\}, \underbrace{J \leq I+1 \wedge I<J \wedge \neg(I+1<J)}_{J \leq I+1 \wedge I<J}\rangle$

$\downarrow$

$\langle \{i \mapsto I+1, j \mapsto J, k \mapsto I\}, J \leq I+1 \wedge I<J\rangle$

SAT[$I \mapsto 2.5, J \mapsto 3$]

do

$\langle S_{2.2.1}; S_{2.2.2}; S_{2.2}; \{i \mapsto I+1, j \mapsto J, k \mapsto I\}, \underbrace{J \leq I+1} \wedge I<J \wedge \underbrace{I+1<J}\rangle$

UNSAT

b)

Sym Exec terminiert muss

2 Pkt Schätzung

$E(x) = \frac{a+b}{2}$

$S(x) = \frac{b-a}{6}$

3 Pkt schätzung

$E(x) = \frac{a+4 \cdot c+b}{6}$

$S(x) = \frac{b-a}{6}$

a: min	c: Mittel	b: max	2.Pkt E(x)	3.Pkt E(x)	S(x)
10	40	60	$\frac{10+60}{2}$	$\frac{10+4 \cdot 40+60}{6}$	$\frac{60-10}{6}$
30	80	160	$\frac{30+160}{2}$	$\frac{30+4 \cdot 80+160}{6}$	$\frac{160-30}{6}$
4	8	12	$\frac{4+12}{2}$	$\frac{4+4 \cdot 8+12}{6}$	$\frac{12-4}{6}$
8	10	20	$\frac{8+20}{2}$	$\frac{8+4 \cdot 10+20}{6}$	$\frac{20-8}{6}$

⊙ 2Pkt Schätzung

$E_{gesamt} = \frac{10+60}{2} + \frac{30+160}{2} \dots = 152$ h

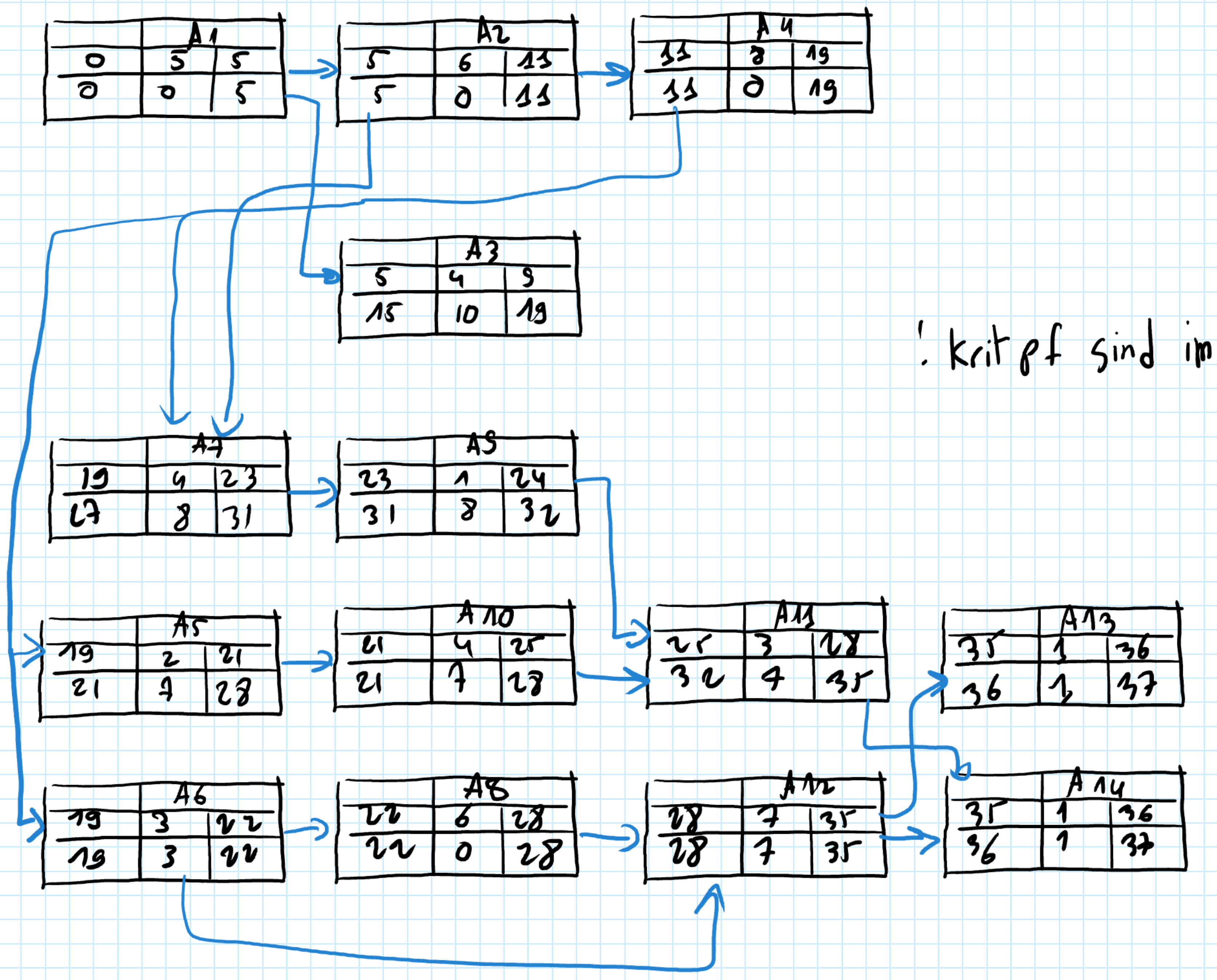
$S_{gesamt} = \sqrt{\left(\frac{60-10}{6}\right)^2 + \left(\frac{160-30}{6}\right)^2 + \dots} \approx 23{,}33$ h

⊙ 3 Pkt Schätzung

$E_{gesamt} = \frac{10+4 \cdot 40+60}{6} + \dots \approx 142{,}66$ h

$S_{gesamt} = \sqrt{\left(\frac{60-10}{6}\right)^2 + \left(\frac{160-30}{6}\right)^2 + \dots} \approx 23{,}33$

Aufgabe 2:



	A1	
0	5	5
0	0	5

	A2	
5	6	11
5	0	11

	A4	
11	8	19
11	0	19

	A3	
5	4	9
15	10	19

	A7	
19	4	23
27	8	31

	A9	
23	1	24
31	8	32

	A5	
19	2	21
21	7	28

	A10	
21	4	25
21	7	28

	A11	
25	3	28
32	4	35

	A13	
35	1	36
36	1	37

	A6	
19	3	22
19	3	22

	A8	
22	6	28
22	0	28

	A12	
28	7	35
28	7	35

	A14	
35	1	36
36	1	37

! krit pf sind immer Aktionen

CP: A1, A2, A4, A6, A8, A12, A14.